



Certified Penetration Testing Professional

hping3 Cheat Sheet

hping3

Source: <http://hping.org>

hping is a command-line oriented TCP/IP packet assembler/analyzer for the TCP/IP protocol that sends ICMP echo requests and supports TCP, UDP, ICMP, and raw-IP protocols.

Syntax

hping3 [options] <Target>

The following table lists the various Hping commands and their respective scanning methods.

1. [Scan Commands](#)
2. [IP Related Commands](#)
3. [TCP/UDP Related Commands](#)
4. [TCP Flags](#)
5. [ICMP Related Commands](#)
6. [Common hping Commands](#)
7. [Protocol Selection Commands](#)

1. Scan Commands

hping Command	Description
hping3 -q <Target> (or) hping3 --quiet <Target>	Scans quietly
hping3 -I <Target> (or) hping3 --interface <Target>	Interface name (to be used for sending packets)
hping3 -D <Target> (or) hping3 --debug <Target>	Debugging info
hping3 -c <Target> (or) hping3 --count <Target>	Packet count (to be sent)
hping3 -V <Target> (or) hping3 --verbose <Target>	Enable verbose output Example: hping3 -V -S <target ip>
hping3 -i <Target> (or) hping3 --interval <Target>	Wait (uX for X microseconds, for example -i u1000)
hping3 -v <Target> (or) hping3 --version <Target>	Show version information and API used to access to data link layer, linux sock packet or libpcap
hping3 -n <Target> (or) hping3 --numeric <Target>	Numeric output
hping3 -z <Target> (or) hping3 --bind <Target>	Used to bind. Use ctrl+z to increment TTL
hping3 -Z <Target> (or) hping3 --unbind <Target>	Used to unbind
hping3 --beep <Target>	Beep for every matching packet received
hping3 --flood <Target>	Sent packets as fast as possible. Don't show replies
hping3 --fast <Target>	Sends 10 packets / sec
hping3 --faster <Target>	Sends 1 packet / μ s

2. IP Related Commands

hping Command	Description
hping3 -a <Target> (or) hping3 --spoof <Target>	Spoof source address
hping3 -A <Target> -p 80	ACK scanning on port 80
hping3 --rand-dest <Target>	Random destination address mode Example: hping3 -1 10.0.1.x --rand-dest -I eth0
hping3 --rand-source <Target>	Random source address mode
hping3 -t <Target> (or) hping3 --ttl <Target>	Set TTL (time to live) value
hping3 -N <Target> (or) hping3 --id <Target>	id [default random]
hping3 -H -ipproto <Target>	Set the IP protocol field, only in RAW IP mode
hping3 -W <Target> (or) hping3 --winid <Target>	Use win* id byte ordering
hping3 -r <Target> (or) hping3 --rel <Target>	Relativize id field to estimate host traffic
hping3 -f <Target> (or) hping3 --frag <Target>	Split packets in more fragments
hping3 -x <Target> (or) hping3 --morefrag <Target>	Set more fragments flag
hping3 -y <Target> (or) hping3 --dontfrag <Target>	Set don't fragment flag
hping3 -g <Target> (or) hping3 --fragoff <Target>	Set fragment offset value
hping3 -G <Target> (or) hping3 --rroute <Target>	Includes RECORD_ROUTE option and display the route buffer
hping3 -m <Target> (or) hping3 --mtu <Target>	set virtual MTU, implies --frag if packet size > MTU

<code>hping3 -o <Target> (or)</code> <code>hping3 --tos <Target></code>	Type of service (default 0x00)
<code>hping3 --lsrr <Target></code>	Loose source routing and record route
<code>hping3 --ssrr <Target></code>	Strict source routing and record route

3. TCP/UDP Related Commands

hping Command	Description
<code>hping3 -s <Target> (or)</code> <code>hping3 --baseport <Target></code>	Base source port [random], +1 on received
<code>hping3 -p <Target> (or)</code> <code>hping3 --destport <Target></code>	Destination port [0] if have, have: +port increased for each reply ++port increased for each sent
<code>hping3 -w <Target> (or)</code> <code>hping3 --win <Target></code>	Set Win size [64]
<code>hping3 -O <Target> (or)</code> <code>hping3 --tcpoff <Target></code>	Set fake TCP data offset
<code>hping3 --keep <Target></code>	Still source port
<code>hping3 -b <Target> (or)</code> <code>hping3 --badcksum <Target></code>	Send packets with a bad IP checksum
<code>hping3 -M <Target> (or)</code> <code>hping3 --setseq <Target></code>	Set TCP sequence number
<code>hping3 -L <Target> (or)</code> <code>hping3 --setack <Target></code>	Set TCP ack
<code>hping3 -Q <Target> (or)</code> <code>hping3 --seqnum <Target></code>	Collects initial sequence number Example: <code>hping3 <target ip> -Q -p 139 -s</code>

4. TCP Flags

hping Command	Description
<code>hping3 -R <Target> (or)</code> <code>hping3 --rst <Target></code>	Set Reset flag Example: <code>hping3 -rst -S <target ip></code>

<code>hping3 -A <Target> (or)</code> <code>hping3 --ack <Target></code>	Set ACK flag Example: <code>hping3 -A <target ip> -p 80</code>
<code>hping3 -F <Target> -p 80 (or)</code> <code>hping3 --fin <Target> -p 80</code>	Set FIN flag scan on port 80 Example: <code>hping3 -F -P -U <target ip> -p 80</code>
<code>hping3 -S <Target> (or)</code> <code>hping3 --syn <Target> -p 80</code>	Set SYN flag Example: <code>hping3 -S 192.168.1.1 -a 192.168.1.254 -p 22 --flood</code>
<code>hping3 -P <Target> -p 80(or)</code> <code>hping3 --push <Target> -p 80</code>	Set PUSH flag scan on port 80 Example: <code>hping3 -F -P -U <target ip> -p 80</code>
<code>hping3 -U <Target> -p 80 (or)</code> <code>hping3 --urg <Target> -p 80</code>	Set URG flag scan on port 80 Example: <code>hping3 -F -P -U <target ip> -p 80</code>
<code>hping3 -X <Target> (or)</code> <code>hping3 --xmas <Target></code>	Set X unused flag (0x40)
<code>hping3 -Y <Target> (or)</code> <code>hping3 --ymas <Target></code>	Set Y unused flag (0x80)

5. ICMP Related Commands

hping Command	Description
<code>hping3 -C <Target> (or)</code> <code>hping3 --icmptype <Target></code>	Set icmp type, default is ICMP echo request (implies --icmp).
<code>hping3 -K <Target> (or)</code> <code>hping3 --icmpcode <Target></code>	Set icmp code, default is 0 (implies --icmp)
<code>hping3 --icmp-ipver <Target></code>	Set IP version of IP header contained into ICMP data, default is 4
<code>hping3 --icmp-iphlen <Target></code>	Set IP header length of IP header contained into ICMP data, default is 5 (5 words of 32 bits)
<code>hping3 --icmp-iplen ip <Target></code>	Set IP packet length of IP header contained into ICMP data, default is the real length
<code>hping3 --icmp-ipid <Target></code>	Set IP id of IP header contained into ICMP data, default is random
<code>hping3 --icmp-ipproto <Target></code>	Set IP protocol of IP header contained into ICMP data, default is TCP.

<code>hping3 --icmp-cksum <Target></code>	Set ICMP checksum, for default is the valid checksum.
<code>hping3 --icmp-ts <Target></code>	Alias for --icmp --icmptype 13 (ICMP timestamp)
<code>hping3 --icmp-addr <Target></code>	Alias for --icmp --icmptype 17 (ICMP address subnet mask)
<code>hping3 --icmp-gw <Target></code>	Set gateway address for ICMP redirect (default 0.0.0.0)
<code>hping3 --icmp-help <Target></code>	Display help for others ICMP options
<code>hping3 --force-icmp <Target></code>	Send all ICMP types (default send only supported types)

6. Common hping Commands

hping Command	Description
<code>hping3 -E <Target> (or)</code> <code>hping3 --file <Target></code>	File inserted into packet's data
<code>hping3 -e <Target> (or)</code> <code>hping3 --sign <Target></code>	Add 'signature'
<code>hping3 -d <Target> (or)</code> <code>hping3 --data <Target></code>	Data size of packet body size
<code>hping3 -j <Target> (or)</code> <code>hping3 --dump <Target></code>	Dump received packets in hex
<code>hping3 -J <Target> (or)</code> <code>hping3 --print <Target></code>	Print dump in printable character
<code>hping3 -B <Target> (or)</code> <code>hping3 --safe <Target></code>	Safe lost packets will be resent
<code>hping3 -u <Target> (or)</code> <code>hping3 --end <Target></code>	Tells when --file reached EOF and prevent rewind
<code>hping3 -T <Target> (or)</code> <code>hping3 --traceroute <Target></code>	Traceroute mode
<code>hping3 --tr-keep-ttl <Target></code>	Keeps TTL fixed

<code>hping3 --tr-stop <Target></code>	Exit when receive the first not ICMP in traceroute mode
<code>hping3 --tr-no-rtt <Target></code>	Don't show RTT
<code>hping3 --tcpexitcode <Target></code>	Use last tcp->th_flags as exit code
<code>hping3 --tcp-mss <Target></code>	Enable the TCP MSS option with the given value
<code>hping3 --tcp-timestamp <Target></code>	Enable the TCP timestamp option to guess the HZ/uptime Example: <code>hping3 -S <target ip> -p 80 --tcp-timestamp</code>
<code>hping3 --apd-send <Target></code>	Send the packet described with APD (see docs/APD.txt)

7. Protocol Selection Commands

hping Command	Description
<code>hping3 -0 <Target> (or)</code> <code>hping3 --rawip <Target></code>	Raw IP mode
<code>hping3 -1 <Target> (or)</code> <code>hping3 --icmp <Target></code>	ICMP mode Example: <code>hping3 -1 <target ip></code>
<code>hping3 -2 <Target> (or)</code> <code>hping3 --udp <Target></code>	UDP mode Example: <code>hping3 -2 <target ip> -p 80</code>
<code>hping3 -8 <Target> (or)</code> <code>hping3 --scan <Target></code>	Scan mode Example: <code>hping3 --scan 1-30,70-90 -S <target website></code> Example: <code>hping3 -8 50-60 -S <target ip> -V</code>
<code>hping3 -9 <Target> (or)</code> <code>hping3 --listen <Target></code>	Listen mode Example: <code>hping3 -9 HTTP -I eth0</code>